

Which a_0 ? The Coefficient of the Radial-Acceleration Relation Under Full Nuisances: A Population-Split Answer

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Abstract

The MOND acceleration scale is conventionally quoted as $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, while a Λ -derived reading gives $a_0 = c^2 \sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11}$ — a 22% (0.11 dex) fork that decades of radial-acceleration-relation (RAR) statistics have failed to close because the coefficient is degenerate with the stellar mass-to-light ratio Y . The standing defense of the lower value — that naive fits omit the per-galaxy distance and inclination nuisances known to be load-bearing — is testable, and this note runs the test: a full hierarchical fit of the SPARC sample (Li et al. 2018 methodology) on the Λ -reading’s own interpolation $v(y) = \sqrt{(1+1/y)}$, with per-galaxy Gaussian priors on distance and inclination from the SPARC master table, a lognormal Y prior, a profiled intrinsic scatter, and validation gates that reproduce both the published McGaugh- v result ($a_0 = 1.20 \times 10^{-10}$ recovered at the published configuration) and the fixed- Y anchor (0.108 dex at $Y = 0.70$). The fit was adversarially verified, including one caught and corrected likelihood error (an inclination-transformation Jacobian) whose correction *strengthened* the adverse result. Four findings, reported at equal weight. **(1) The full-sample tension is real.** With nuisances properly marginalized, the full-SPARC optimum on the Λ -reading’s own v is $a_0 = 1.24\text{--}1.37 \times 10^{-10}$, and 9.36×10^{-11} is excluded at $>3\sigma$ -equivalent on every honest error treatment (naive $\Delta\chi^2$ 161–270; galaxy-bootstrap $\approx 3.3\sigma$; leave-out-most-influential holds; a global stellar-population zero-point cannot rescue it). The missing-nuisance defense is closed: the nuisances are in, and the tension survives them. **(2) But no single a_0 fits SPARC on any interpolation.** Gas-dominated and star-dominated sub-populations prefer coefficients differing by a factor ≈ 1.8 — on the Λ -reading’s v *and* on McGaugh’s — so the “measured a_0 ” is entangled with population-level baryon systematics by construction, for every party to the dispute. **(3) The only mass-to-light-free measurement lands on the Λ value.** In gas-dominated galaxies Y is dynamically irrelevant, making their preferred a_0 the cleanest coefficient statement the data can produce; with full nuisances it is 9.8×10^{-11} — statistically centered on $c^2 \sqrt{(\Lambda/32\pi)}$ ($\Delta\chi^2(9.36 \times 10^{-11}) \leq 1.5$ under both likelihood conventions). An earlier low-side preference (7.7×10^{-11}) from a simplified fit is retracted here as an artifact of the corrected Jacobian term. **(4) The high-side preferences cohere as one systematic.** The star-dominated SPARC preference and the independent KiDS weak-lensing preference sit a *common* $\approx 1.4\text{--}1.6$ baryon uplift from the Λ value (+0.146 dex in Y ; +0.20 dex in lensing baryons) — one coherent shift, in the two regimes where baryon accounting is least certain, rather than independent per-dataset failures. The honest conclusion is not a verdict but a diagnosis: **on current data the RAR coefficient is not a single measurable number.** The mass-to-light-free evidence centers on the Λ -derived value; the star-dominated bulk demands $\approx 1.4 \times 10^{-10}$ or a coherent baryon uplift; and the deciding measurements are per-population baryon physics — independent bulge mass-to-light ratios and absolute baryon budgets — not more RAR statistics.

1. The fork, and why statistics alone never closed it

In the deep-MOND regime $g_{\text{obs}} \approx \sqrt{(g_{\text{bar}} a_0)}$, so a rescaling of the baryonic mass ($Y \rightarrow \lambda Y$) and of the coefficient ($a_0 \rightarrow a_0/\lambda$) move the prediction identically: the RAR constrains the *product*, and the quoted a_0 is a convention anchored to an assumed stellar population scale. This is why the same data support $a_0 = 1.2 \times 10^{-10}$ at $Y_{\text{disk}} \approx 0.5$ and $a_0 = 9.36 \times 10^{-11}$ at $Y_{\text{disk}} \approx 0.70$ with nearly identical scatter (0.108 dex vs 0.122 dex on the respective v 's). Two things can break the degeneracy: nuisance-level information (distances and inclinations shift radii and velocities differently than Y shifts masses), and — more decisively — galaxies where Y does not matter.

2. Method and gates

We fit all SPARC rotation-curve points (171 usable galaxies) with the Λ -reading's interpolation, treating per galaxy: distance D (Gaussian prior, catalog value $\pm$ its uncertainty; radii scale as D , component velocities as $\sqrt{D}$), inclination i (Gaussian prior; observed velocities rescale as $\sin i_0/\sin i$), and Y_{disk} (lognormal, 0.5 ± 0.1 dex; $Y_{\text{bul}} = 1.4 Y_{\text{disk}}$), with a global intrinsic scatter profiled. Validation gates: (i) the published configuration (McGaugh v , $Y = 0.5$, fixed D , i) reproduces $a_0 = 1.20 \times 10^{-10}$; (ii) the fixed- Y anchor reproduces the committed 0.108 dex. An independent adversarial verification re-derived the load-bearing numbers with separately written code, and caught one real error — the likelihood's inclination-rescaled errors carried a normalization term without its data-transformation Jacobian, biasing a_0 low. All headline numbers below use the corrected (Li-faithful, data-space) likelihood; the uncorrected variant is reported in the repository as a labeled fork. Because rotation-curve points within a galaxy share systematics, naive $\Delta\chi^2$ overstates significance; galaxy-level bootstrap values are quoted alongside throughout.

3. The full-sample verdict: the tension is real

With the nuisances marginalized, the full-sample optimum on the Λ -reading's own v is $\mathbf{a_0} = 1.37 \times 10^{-10}$ (1.24×10^{-10} on the uncorrected fork), with 9.36×10^{-11} disfavored at $\Delta\chi^2 = 270$ naive (161 on the fork), **$P < 0.002$ ($\approx 3.3\sigma$) by galaxy bootstrap**, robust to removing the two most influential galaxies, and *not* rescuable by a global Y zero-point shift within its plausible band. We state this at full weight: the long-standing defense that the tension was an artifact of missing distance/inclination nuisances is now closed, because the nuisances are included and the tension survives. Three caveats keep it short of a kill, and they are structural rather than statistical: the bulge mass-to-light ratio is locked to the disk's ($Y_{\text{bul}} = 1.4 Y_{\text{disk}}$) while the most influential galaxies are bulge-heavy — an independent- Y_{bul} fit is the one unrun knob; the intrinsic-scatter model is a lever; and the population split of §4 means the “full-sample a_0 ” is an average over sub-populations that genuinely disagree.

4. No single a_0 fits SPARC — on any v

Splitting the sample by dynamical gas fraction, the gas-dominated and star-dominated populations prefer coefficients differing by a factor ≈ 1.8 , and the split persists on McGaugh's interpolation as well as the Λ -reading's. Whatever one's theoretical commitments, this is a model-independent statement about the data: **the RAR coefficient extracted from SPARC is population-dependent at the factor-two level**, and every published single-number a_0 —

1.2×10^{-10} included — is an average over that disagreement. Any honest use of “the measured a_0 ” against (or for) a theoretical value must carry this entanglement.

5. The mass-to-light-free measurement

Gas-dominated galaxies are the exception to the degeneracy of §1: their baryonic mass is set by HI, not starlight, so their preferred coefficient is insensitive to Y — the cleanest coefficient measurement the data contain (the same logic that made gas-rich galaxies the classic honest test of the baryonic Tully–Fisher relation; McGaugh 2011). With full nuisances and the corrected likelihood, the gas-dominated cut ($N = 37$) prefers $a_0 = 9.8 \times 10^{-11}$, **statistically centered on the Λ -derived 9.36×10^{-11}** ($\Delta\chi^2 \leq 1.5$ under both conventions; bootstrap $P(\leq 9.36 \times 10^{-11}) = 0.39$ — i.e., the Λ value is entirely typical). We flag prominently that a preliminary version of this cut, run without the distance/inclination machinery, preferred 7.7×10^{-11} and was briefly recorded as a “low-side” hint; the adversarial verification traced that sign to the Jacobian error, and it is retracted and superseded here. The corrected, convention-robust statement stands: **where the mass-to-light ratio cannot act, SPARC lands on $c^2\sqrt{(\Lambda/32\pi)}$** .

6. The high side coheres as one uplift

The measurements that prefer the high coefficient are precisely those where baryon accounting carries the most freedom, and they miss the Λ value by a *common* factor: the star-dominated SPARC population by a Y uplift to ≈ 0.70 (+0.146 dex over the population-synthesis mean), and the independent KiDS-1000 weak-lensing RAR by a +0.20 dex baryon uplift (with the papers’ own circumgalactic-gas accounting identified as the one-sided systematic pointing that way). One coherent $\times 1.4$ – 1.6 uplift across two independent datasets and techniques is a different epistemic object than two unrelated misses — it is either a real, physically-identified baryon undercount (in which case the Λ value survives everywhere), or it is the Λ -reading’s coherent failure mode (in which case the high value wins everywhere). Resolved stellar-mass measurements in gas-rich dwarfs (MIGHTEE-HI) currently measure low Y for *that* population without transferring to SPARC’s star-dominated spirals — consistent with either reading of the uplift.

7. What actually decides it

More RAR statistics will not: the degeneracy structure guarantees it. The deciding measurements are: (i) **independent bulge mass-to-light ratios** for the influential bulge-heavy galaxies (unlocking Y_{bul} is the single unrun knob in the hierarchical fit); (ii) **absolute baryon budgets** — pinning the stellar-population zero-point and the circumgalactic contribution at the 0.05-dex level, which adjudicates the uplift directly; (iii) **larger gas-dominated samples** (the forthcoming ~ 4000 -galaxy BIG-SPARC will multiply the Y -free cut tenfold); and (iv) any external, non-RAR measurement of a_0 -scale physics. Until then the honest summary is the population-split one, and this note commits to it symmetrically: if independent bulge and baryon measurements confirm the population-synthesis values, the full-sample tension of §3 becomes a genuine coefficient exclusion and the Λ reading fails; if they confirm the coherent uplift, the Λ reading survives its sharpest current test with its cleanest evidence already in hand.

Scope and credit

The hierarchical methodology is Li et al. (2018); the data are SPARC (Lelli, McGaugh & Schombert 2016); the gas-rich logic is McGaugh’s; the lensing comparison rests on Brouwer et al. (2021) and Mistele et al. (2024); the resolved-Y context is MIGHTEE-HI (Văršeteanu et al. 2025). This note’s contributions are the arbiter run itself on the Λ -reading’s own interpolation, the corrected-likelihood verification, the population-split diagnosis, the Y-free centering result, and the coherent-uplift framing. The Λ -derived coefficient is a one-parameter modelling choice whose earlier over-claims were publicly retracted (2026-06-23); nothing here derives it, and §3’s adverse finding is banked in the same public ledger as §5’s favorable one.

References

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Reproducible: `reviews/inhand_2026_07/{lane_L1_hierarchical_sparc_arbiter, VERIFY_L1_independent, VERIFY_L1_inclination_norm_fork}.py` (all exit 0; the uncorrected-likelihood fork preserved and labeled). Companion results ledger: `INHAND_CALCS_RESULTS_2026-07.md`. The author retracted all earlier “theory of everything” claims (2026-06-23); this note makes none.